

Aptitude Day-6

1. In how many ways can 3 student be selected from 7 students?

Soln:

3 student from 7

$${}^7C_3 = \frac{7!}{3! \times 4!}$$

$$= \frac{7 \times 6 \times 5}{3 \times 2 \times 1}$$

$$= 7 \times 6 \times 5 = 210$$

$$= 3 \times 2 \times 1 = 6$$

$$= 210 \div 6$$

$$= 35.$$

2) Soln:

2 Digit number using 1, 2, 3, 4

4 choice $\rightarrow (1, 2, 3, 4)$

3 choice $\rightarrow$ (remaining digit)

$$= 4 \times 3 = 12$$

$$= 12.$$

3) Soln:

Captain & Vice-Captain from 6 player

choice 6 Captain

choice 5 Captain

$$= 6 \times 5 = 30$$

$$= 30.$$

Soln:

Arrange letters of "MATH"

Total letters = 4

1st $\rightarrow$ 4 choice

2nd $\rightarrow$ 3

3rd $\rightarrow$ 2

4th $\rightarrow$ 1

choice H $\div$ each other

$$4 \times 3 =$$

$$= 4 \times 3 \times 2 \times 1 = 24$$

$$= 24.$$

5). soln:

Committee of 4 from 8 people

$${}^8C_4 = \frac{8!}{4! \times 4!}$$

$$= \frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2 \times 1}$$

$$= 8 \times 7 \times 6 \times 5 = 1680$$

$$= 4 \times 3 \times 2 \times 1 = 24$$

$$= 1680 \div 24$$

$$= 70.$$